Authenticity Matters: Enhancing Product Consistency via Localized Optimization in AI-Generated Sponsored Content

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Abstract

Recent text-image-to-video diffusion models enable marketers to produce short-form sponsored videos at scale, but their global denoising objectives do not preserve consistency in locally salient regions, particularly where the promoted product appears. In sponsored content, inaccuracies in product packaging, branding, and on-pack text undermine advertising authenticity, weaken consumer trust, and may trigger platform enforcement. This paper makes two contributions. First, we propose a localized similarity optimization framework that augments the diffusion training objective with a region-of-interest (ROI) weighting placing greater optimization mass on product voxels detected within each frame, using product masks obtained automatically from a catalog image and an open-vocabulary detector. Second, we report a diagnostic empirical study on a large company-sponsored dataset of TikTok shoppable videos that motivates the proposed framework. Fine-tuning Wan2.2-TI2V-5B under five conditions, we find that (i) image-only LoRA conditioning (FVD $= 200.37$) outperforms multimodal LoRA conditioning that adds the textual script (227.72), and (ii) switching from LoRA to full-parameter multimodal fine-tuning closes only a small fraction of this gap (206.45) and still does not match the image-only ceiling. Adding modalities and adding adaptation capacity therefore do not address the product consistency bottleneck, indicating that the bottleneck is structural rather than a matter of model capacity, and motivating the region-aware training objective that we propose. Empirical validation of the ROI loss itself is left to future work; this paper establishes the diagnosis and the proposal.

1 Introduction

Generative AI is transforming how marketing content is created and deployed. Vision–language diffusion models can produce short-form videos from a textual prompt and a reference product image, dramatically lowering production cost for sponsored content on shoppable platforms such as TikTok. Yet marketers and platforms increasingly observe that AI-generated sponsored videos fail to preserve the authenticity of the advertised product: shape, packaging, color, and on-pack text exhibit subtle but visible deviations from the actual product.

Authenticity is a well-established driver of trust, brand attitude, and purchase intention [15, 13]. In shoppable short-video commerce, the most diagnostic authenticity cue is the visual representation of the product itself; small deviations both reduce credibility [7] and trigger automatic platform enforcement under marketplace integrity policies. Maintaining product fidelity in AI-generated sponsored content is therefore both a marketing and a compliance requirement.

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State-of-the-art video diffusion models, however, optimize a global denoising loss that averages alignment across the entire visual field. This implicitly treats all spatial regions as equally important, while in fact the informational and economic importance of regions in advertising is highly heterogeneous: viewers attend disproportionately to the promoted product [16]. The product region is also typically the structurally complex foreground (logos, text, packaging shape) that is hardest for a diffusion model to reproduce, while easier-to-render background voxels absorb gradient capacity. As a result, generated videos may appear globally coherent yet contain localized inaccuracies that undermine authenticity.

We argue that this misalignment between current generative objectives and marketing relevance can be addressed at the training-objective level. We propose a localized similarity optimization framework that places greater optimization mass on voxels inside the product region. We derive a per-frame product mask from a catalog image and an open-vocabulary detector [5], downsample the mask to the latent space of the 3D VAE used by the diffusion backbone, and train under a single mask-weighted denoising loss. On a large company-sponsored dataset of TikTok shoppable videos, a five-condition diagnostic ablation rules out two natural alternatives to such a region-aware design: adding the textual script as an additional conditioning modality on top of the first-frame image actually *worsens* Fréchet Video Distance under LoRA fine-tuning (from 200.37 to 227.72), and switching from LoRA to full-parameter multimodal fine-tuning closes only a small fraction of this gap (206.45). Neither richer conditioning nor higher adaptation capacity recovers the product consistency that image-only fine-tuning already provides, indicating a structural rather than a capacity bottleneck and motivating the proposed region-aware loss.

Contributions. (i) We identify product accuracy as a measurable design objective for AI-generated sponsored content and propose to operationalize it directly inside the diffusion training loss rather than as a downstream evaluation criterion. (ii) We formulate a mask-weighted denoising loss with a single hyperparameter λ, requiring only an automatically extracted product mask and no auxiliary encoders or multi-objective scheduling. (iii) We report a diagnostic five-condition ablation on a large industry dataset of shoppable short videos that empirically motivates the region-aware design: neither adding modalities nor scaling from LoRA to full-parameter fine-tuning closes the product consistency gap. Empirical validation of the ROI loss itself is left to future work.

2 Related Work

Text-to-video diffusion. Diffusion models have become the dominant generative paradigm for visual content [9], with recent text-to-video and text-image-to-video extensions producing increasingly coherent short-form video. These models typically train under a global denoising loss that does not differentiate among spatial regions. This is well-suited to general visual realism but ill-suited to advertising contexts, where some regions carry far more economic weight than others.

Region-aware generation. A growing literature in computer vision incorporates spatial structure into diffusion training and inference. Examples include explicit region grounding using bounding boxes or masks [4, 12, 19, 20], localized attention guidance during inference [8, 3, 21], and patch- or panel-based composition frameworks [1]. Most relevant to us, region-adaptive compression with spatially varying diffusion [17] establishes that different spatial regions can be assigned different reconstruction priorities within a single diffusion objective. We adapt the spatially varying weighting idea from compression to advertising authenticity, encoding marketing-relevant priorities directly into the training loss.

Authenticity in marketing and AI-generated content. Marketing research treats authenticity as a multidimensional consumer judgment that drives trust and persuasion [15, 13, 11]. As AI systems begin to author sponsored content, recent work emphasizes the need to preserve the authenticity signals consumers rely on [14, 6], but offers limited guidance on how to encode such signals at the training-objective level. Empirical operationalizations have largely focused on perception measures [15], manual coding [2], or linguistic indicators [11, 18], which transfer poorly to short-form video. Our work bridges these literatures by treating product authenticity as a measurable, region-level design objective for video diffusion training.

3 Method

3.1 Overview

We build on the open-source DiffSynth-Studio diffusion engine and the Wan2.2-TI2V-5B text-image-to-video backbone, and extend its standard denoising objective with a region-of-interest (ROI) weighting. The pipeline has three steps: (i) a multimodal preprocessing stage that produces, for each shoppable training video, a structured textual script and a per-frame product mask; (ii) a latent alignment stage that downsamples the pixel-space mask to match the latent video tensor; (iii) low-rank fine-tuning of the backbone under a mask-weighted denoising loss.

3.2 Data Preprocessing

Each training instance pairs a shoppable short video with a high-resolution catalog image of the promoted product. We construct two preprocessing artifacts.

Script extraction. We feed each video and its audio track to Qwen3-Omni, a multimodal vision–language model, which returns a structured script containing (i) a product description summarizing the product's category, key visual attributes, and use case and (ii) a sequence of scene-level entries, each consisting of a spoken-text description and a visual description of the on-screen content. This script becomes the textual prompt that conditions the backbone at training and inference time.

Product region detection. We localize the product within each frame using YOLOWorld [5], an open-vocabulary object detector queried with the catalog image and product description. For each frame in which the product is present, the detector returns the standard four-dimensional bounding-box parameterization

$$b_t = (c_t^{(x)},\ c_t^{(y)},\ w_t,\ h_t) \in [0,1]^4, \tag{1}$$

where $(c_t^{(x)}, c_t^{(y)})$ is the center point of the box in normalized image coordinates and (w_t, h_t) are its normalized width and height. We use b_t in two complementary ways. First, the four numbers themselves form a per-frame control signal that informs the backbone, frame by frame, both *where* the product is expected to appear (via the center point) and *how large* it is expected to be (via the width and height); stacked along time, the boxes give a control trajectory $B \in [0,1]^{T\times 4}$. Second, we rasterize each b_t into a binary mask $M^{\text{pixel}} \in \{0,1\}^{T\times H\times W}$, where T, H, W are the temporal, vertical, and horizontal dimensions of the reference video; voxels inside the product box take value one and voxels outside take zero. The control trajectory and the rasterized mask thus capture, respectively, the spatial location and scale of the product (sparse) and its spatial extent (dense), the latter of which drives the mask-weighted loss in Eq. (3).

3.3 Mask Construction and Latent Alignment

The Wan2.2-TI2V-5B backbone operates on a latent video produced by a 3D VAE that downsamples the input by a factor of 4 in time and 16 in each spatial dimension. To apply the pixel-space product mask to the latent denoising target, we downsample M^{pixel} to the latent resolution using nearest-neighbor interpolation, yielding a latent-space mask

$$M \in \{0,1\}^{t\times h\times w}, \qquad t = T/4,\ \ h = H/16,\ \ w = W/16, \tag{2}$$

that has the same spatial–temporal shape as the latent video tensor $z_t \in \mathbb{R}^{t\times h\times w\times c}$. Because the VAE encoder and the mask downsampling share the same factors, the latent mask remains spatially aligned with the latent representation of the reference video.

3.4 Mask-Weighted Diffusion Training

We replace the uniformly weighted denoising loss with a mask-weighted variant in which voxels inside the product region receive weight $1+\lambda$ and background voxels receive weight 1:

$$\mathcal{L} = \mathbb{E}_{x,\epsilon,t}\Big[\, \big\| \,(1+\lambda\cdot M)\odot\big(\epsilon - \epsilon_\theta(z_t,t)\big)\,\big\|_2^2 \,\Big], \tag{3}$$

where z_t is the noisy latent video at timestep t, ϵ is Gaussian noise from the forward diffusion process, ϵ_θ is the noise prediction produced by the backbone, $\odot$ is elementwise multiplication, and $\lambda \geq 0$ is a

non-negative scalar controlling the additional weight on the product region. When $\lambda = 0$, Eq. (3) reduces to the standard global denoising loss.

The mask-weighted formulation has three desirable properties. First, the gradient signal in the product region is amplified relative to the background, encouraging the model to allocate representational capacity toward visual attributes that determine product authenticity. Second, background voxels continue to receive non-zero gradients, ensuring that the generated video remains globally coherent. Third, the loss reduces to a single scalar that integrates seamlessly into the existing diffusion training pipeline; no auxiliary encoder or external alignment module is required at training time, which avoids the instability that often accompanies multi-objective losses. The objective is intended to be optimized via low-rank adaptation [10] or full-parameter fine-tuning of the Wan2.2-TI2V-5B backbone.

Empirical motivation and scope. The empirical study reported in Section 4 establishes the diagnosis that motivates this objective. Specifically, we show that two natural alternatives to a region-aware design — adding the textual script as an extra conditioning modality, and switching from LoRA to full-parameter fine-tuning — both fail to close the product consistency gap relative to image-only fine-tuning, indicating that the bottleneck is structural rather than a matter of conditioning richness or adaptation capacity. A full empirical evaluation of the proposed ROI loss on the same conditions, including sweeps over λ and product categories, is left to future work; the present paper establishes the diagnosis and the proposed remedy.

4 Empirical Analysis: Modality and Capacity Are Not Enough

This section reports a diagnostic empirical study whose purpose is to motivate the region-aware training objective proposed in Section 3. We ask two questions. First, does adding the textual script as an extra conditioning modality on top of the first-frame product image improve product consistency over image-only conditioning? Second, if it does not, is the gap merely an artifact of low-rank adaptation, in the sense that giving the model more trainable parameters via full-parameter fine-tuning would close it? Answering both questions in the negative provides empirical motivation for moving beyond modality and capacity scaling toward a structurally region-aware objective.

4.1 Setup

Backbone and generation settings. We fine-tune Wan2.2-TI2V-5B and generate videos at resolution 832×480 ($H \times W$) with 121 frames at 16 FPS (approximately 7.5 seconds). We use a per-step batch size of one video. Under LoRA fine-tuning (rank 64), one epoch over the training subset takes approximately 70 minutes on a single GPU; full-parameter fine-tuning is correspondingly more expensive but uses the same data, optimizer schedule, and number of epochs to keep the comparison interpretable.

Dataset. We use a company-sponsored dataset from a US-based multi-channel network operating primarily on TikTok. The dataset spans April 1, 2024 to June 30, 2025 and contains roughly $339{,}819$ unique sponsored shoppable videos produced by $1{,}833$ creators promoting $70{,}357$ products across $22{,}238$ brands and 27 top-level / 207 second-level product categories (e.g., Beauty & Personal Care, Womenswear, Sports & Outdoor, Kitchenware, Food & Beverages). Each video is paired with the catalog image of the promoted product, which provides the reference image and the product description used to prompt the model.

4.2 Conditions

We compare five conditions that jointly vary input modalities and adaptation capacity. Conditions A–C and the proposed reference point use LoRA adaptation; condition E switches to full-parameter fine-tuning while keeping the same multimodal conditioning as C.

- **(A) Script only (LoRA).** Fine-tuned and inferred using only the Qwen3-Omni script. Isolates the contribution of textual conditioning under LoRA.
- **(B) First Image (LoRA).** Fine-tuned and inferred using only the first-frame product image. Isolates the contribution of visual conditioning under LoRA.

- **(C) Script + First Image (LoRA).** Both modalities under LoRA. Standard multimodal fine-tuning baseline.
- **(D) Without Train.** Pretrained Wan2.2-TI2V-5B used directly without any fine-tuning. Off-the-shelf prompt-based baseline.
- **(E) Script + First Image (full-parameter).** Same multimodal conditioning as C, but the entire backbone is fine-tuned end-to-end rather than via LoRA. Tests whether the C–B gap is a capacity artifact.

The proposed ROI-weighted loss (Eq. (3)) is *not* evaluated as one of these conditions; its empirical evaluation is left to future work, and the role of conditions A–E here is purely diagnostic.

4.3 Metrics

FVD measures the distance between the distributions of generated and reference videos in the feature space of a pretrained video classifier; lower values indicate that generated videos are distributionally closer to the reference shoppable videos. **Per-frame cosine similarity** captures local consistency: for each frame index, we extract embeddings of the generated and reference frames using a pretrained visual encoder and compute their cosine similarity, yielding a similarity trajectory along the time axis. Higher per-frame similarity indicates that, frame by frame, generated content remains visually close to the reference, which is the relevant notion of consistency for product authenticity.

4.4 Results

Distributional fidelity. Table 1 reports FVD across the five conditions. Four observations follow.

(i) Fine-tuning on shoppable video is necessary. The untrained backbone (D) achieves the worst FVD (311.54). Despite the strong general-purpose generation capabilities of Wan2.2-TI2V-5B, off-the-shelf inference is poorly aligned with the visual distribution of TikTok-style shoppable short video.

(ii) Adding the textual script does not help over image-only LoRA. Comparing the LoRA conditions, image-only fine-tuning (B, 200.37) outperforms multimodal fine-tuning (C, 227.72) by 27.35 FVD points, while text-only fine-tuning (A, 269.94) is much worse than both. The first-frame product image is the dominant source of visual signal in this domain. Augmenting it with the script does provide semantic controllability, but it also introduces a measurable distributional drift: in this metric, two modalities are *worse* than the right single modality.

(iii) Full-parameter fine-tuning does not close the C–B gap. Switching from LoRA (C, 227.72) to full-parameter fine-tuning (E, 206.45) on the same multimodal inputs improves FVD by only 21.27 points and remains worse than the image-only LoRA baseline (B, 200.37). The C–B gap therefore cannot be explained as a LoRA capacity artifact; even when every parameter of the backbone is trainable, multimodal fine-tuning still fails to recover the consistency that image-only LoRA already attains.

(iv) Diagnostic implication. The combination of (ii) and (iii) is the central empirical finding: neither richer conditioning nor higher adaptation capacity is sufficient to close the product consistency gap relative to image-only fine-tuning. Conditioning information that is highly relevant for narrative control (the script) systematically pulls the model away from the product region's reference distribution, and this misallocation of training signal persists when the model is given enough parameters to in principle use both modalities optimally. The bottleneck is therefore structural: it concerns where the training signal is allocated within each frame, not how much information is conditioned on or how many parameters are updated.

Per-frame consistency. Figure 1 reports per-frame cosine similarity over the 121-frame horizon. The text-only and untrained conditions ("w/o image" and "w/o train") exhibit consistently low similarity throughout, fluctuating between roughly 0.63 and 0.69. Conditions that include the first-frame image start near 0.95–0.98 at frame zero, where the first frame is essentially anchored to the conditioning image, and decay as the model has to imagine subsequent frames. Both multimodal conditions (LoRA and full-parameter) sustain similarity in the 0.70–0.78 range across the sequence and slightly exceed image-only LoRA ("w/o text") in mid-sequence frames; full-parameter multimodal training maintains a small but consistent advantage over LoRA multimodal training in early frames (frame

Table 1: Fréchet Video Distance across the five diagnostic conditions on the TikTok shoppable video dataset (lower is better). Adding the textual script to the first-frame image (C) is *worse* than image-only fine-tuning (B); switching from LoRA to full-parameter fine-tuning (E) closes only part of this gap and still does not reach the B ceiling. Neither modality nor capacity scaling closes the product consistency gap, motivating the region-aware loss proposed in Section 3.

Condition	**Configuration**	**FVD** (↓)
A	Script only (LoRA)	269.94
B	First Image (LoRA)	**200.37**
C	Script + First Image (LoRA)	227.72
D	Without Train	311.54
E	Script + First Image (full-parameter)	206.45

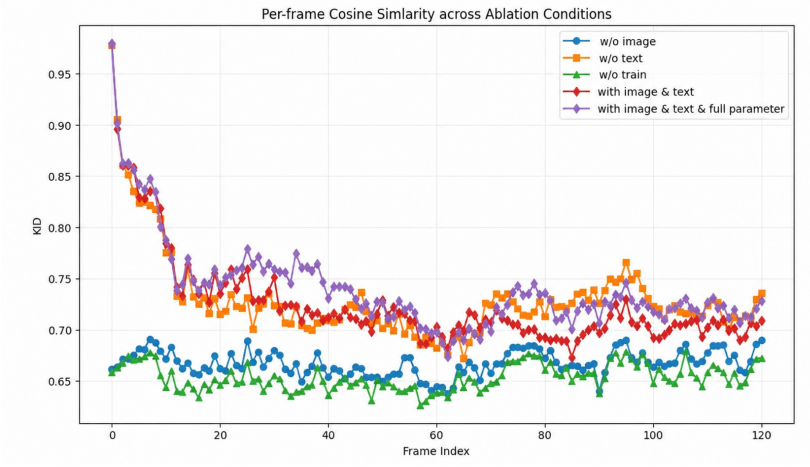

Figure 1: Per-frame cosine similarity between generated and reference frames for the five diagnostic conditions over the 121-frame horizon (higher is better). Multimodal training ("with image & text" and "with image & text & full parameter") provides modest local-consistency gains over image-only fine-tuning ("w/o text") in mid-sequence frames, but does not translate into a distributional improvement at the FVD level (Table 1).

indices 0–30). However, neither multimodal variant translates this per-frame advantage into a corresponding gain at the distributional level: their FVD scores remain above the image-only ceiling. This is consistent with the diagnostic reading above: the script helps locally on frames where it can guide content generation, but the textual conditioning shifts the global distribution of generated frames away from the reference shoppable distribution, and additional adaptation capacity is not sufficient to undo that shift.

Takeaway. Together, the FVD and per-frame results indicate that the central question for AI-generated sponsored video is not "how should we add more information or more parameters?" but "how should we allocate training signal across the visual field?". Image-only LoRA already places most of its training signal on the product region by virtue of the conditioning, while multimodal training spreads its signal more evenly — a redistribution that helps narrative control but degrades product fidelity, and that even full-parameter fine-tuning is not powerful enough to undo. The proposed mask-weighted loss (Eq. (3)) directly addresses this signal-allocation question by re-weighting the diffusion training objective at the voxel level. Establishing whether this region-aware design closes the C–B gap, and how that closure depends on the ROI weight λ, the masked region, and the product category, is the natural next experiment.

5 Discussion and Future Work

Limitations. Three limitations are worth highlighting. First, the proposed ROI loss is not evaluated empirically in this paper; the experiments here are purely diagnostic and the comparison between

condition C, condition E, and a hypothetical ROI-augmented condition has not been run. Second, the diagnostic results reflect a single backbone, a fixed resolution and clip length, and the product distribution of one MCN partner; generalization across backbones, resolutions, and product categories remains to be tested. Third, even when the ROI loss is implemented, the present scoping treats only the product region as the relevant ROI, and the relative weight λ is a free hyperparameter whose effect on product fidelity, background coherence, and category-specific behavior has not been characterized.

Future work. The findings open two directions. (i) *Empirical evaluation of the proposed loss.* The most immediate next experiment is to add a sixth condition — multimodal training with the mask-weighted loss in Eq. (3), under both LoRA and full-parameter fine-tuning — and ask whether it closes the C–B and E–B gaps documented in Table 1. A sweep over λ and over alternative mask sources (open-vocabulary detection, segmentation, soft attention) would characterize the design space. (ii) *Multi-region authenticity.* Sponsored videos often feature a creator presenting the product, and combining the product mask with a face-region mask yields a multi-region weighted loss that captures both product and creator authenticity; the formulation in Eq. (3) extends to any number of region maps.

Contribution Statement

Kunhan designed and implemented the code for the diagnostic experiments reported in Section 4, and contributed the original idea of recasting product authenticity as a region-aware training-objective problem (Sections 1 and 3). Claude was used as a writing assistant to help draft and revise the manuscript.

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